

Syllabus

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Instructor

Chen Ying, Assistant Professor, Teaching Stream

c.ying@utoronto.ca

Teaching Assistants

Yining Wang: yining.wang@mail.utoronto.ca

Yiren Zhao: yiren.zhao@mail.utoronto.ca

Overview

This graduate-level course offers students hands-on experience in developing creative and innovative mobile applications using **React Native** and **Expo**. Students will complete four progressive assignments and a collaborative group project, mastering skills such as cross-platform development, user notifications, state management, and app deployment. By the end of the course, students will build a fully functional mobile app, demonstrating their technical skills and creativity, and preparing them for careers in modern mobile development.

Learning Objectives

By the end of this course, students will be able to:

- Develop cross-platform mobile applications using React Native and Expo with TypeScript, implementing core components, hooks, and UI design principles to create responsive and user-friendly interfaces.
- Implement robust navigation systems for multi-screen mobile apps using React Navigation or Expo Router, incorporating TypeScript for typed navigation and data passing between screens.
- Manage app state and persist data locally using Context API, useReducer, and React Native Async Storage with TypeScript, ensuring seamless user experiences across app sessions.
- Integrate Expo Notifications to implement local or push notifications, enhancing user engagement through timely and interactive alerts.
- Connect mobile applications to backend services or APIs (e.g., REST, Supabase, Firebase) using TypeScript for secure data fetching, dynamic UI updates, and error handling.
- Explore advanced mobile development features, such as user authentication, real-time updates, mobile sensors, or custom animations, to enhance app functionality and creativity.

- Collaborate effectively on a team-based, creative mobile app project, leveraging GitHub for version control and project management, and deploy it using Expo EAS Build for testing on iOS or Android devices.

Prerequisites

Students should have a strong foundation in JavaScript, including proficiency with ES6+ features such as arrow functions, destructuring, and asynchronous programming using `async/await`. Prior experience with React, particularly functional components, props, state, and hooks, is highly recommended. A brief React refresher will be provided in the first lecture. Familiarity with TypeScript is preferred but not required, as it will be used throughout the course with guidance offered as needed.

Weekly Schedule

As this is the first time this course is offered, the topics may be subject to change. The following weekly schedule provides a tentative list of topics that will be covered in this course:

Week	Lecture	Lecture Date	Topic
1	1	September 3	Course Introduction and React Refresher
2	2	September 10	React Native Fundamentals
3	3	September 17	Styling and UI Design
4	4	September 24	Navigation with React Navigation
5	5	October 1	Navigation with Expo Router
6	6	October 8	State Management in React Native
7	7	October 15	Redux Toolkit and Data Persistence
8	8	October 22	Notifications and User Engagement
9	-	-	Reading Week (No Lecture)
10	9	November 5	Backend Integration and User Authentication
11	10	November 12	Deploying Mobile Apps with Expo EAS
12	11	November 19	Course Project Presentations
13	12	November 26	Course Project Presentations

Grading

Assignments (50%)

Each assignment should be completed individually.

- 4 assignments, 12.5% each

Project (50%)

Students will work in groups of 2 to 4 to develop a mobile application implementing the modern technologies learned throughout the course.

- **Proposal** (15%): due on October 19, 2025, 11:59 PM, Sunday
- **Presentation** (10%): November 19 & November 26
- **Final Deliverable and Report** (25%): due on December 7, 2025, 11:59 PM, Sunday

Bonus Points

In-class quizzes will be held at some lectures. If a quiz is scheduled, students will be notified in the previous lecture. Each quiz will only cover material from that lecture.

We will have a quiz in the first lecture.

Course Policies

Academic Integrity

Students are expected to follow the University's [Academic Integrity Policies](#) [↗] and the [Code of Behaviour on Academic Matters](#) [↗]. All submitted work must be original and properly attributed.

Assignment Remarking

An [Assignment Remarking Request Form](#) must be submitted to the instructor **in person on paper** (after a lecture or during office hours) **within two weeks of the grade release**. Email submissions are **not accepted**. If you are unable to attend in person, you may ask a classmate to submit the form on your behalf.

Requests for **grading details or explanations** are considered part of the remarking process, since full grading criteria and test details will be published on the assignment handout after grades are released. Please **do not contact TAs or instructor** for individual grading breakdowns — instead, submit a formal remark request if you would like your work to be re-evaluated.

All remarking requests will be **reviewed together after the two-week submission window closes**. Students will receive a notification once the remarking process has been completed.

This policy ensures **fairness and consistency** across all students and helps manage the TA's workload in a large class, where informal grading discussions can quickly become unscalable.

Communication

A [private Git repository](#) [↗] has been set up specifically for discussion purposes. To gain access to the Git repository, **please submit your GitHub username via [this assignment](#) [↗] on Quercus.**

The [GitHub Discussions](#) [↗] feature will serve as a platform for students to ask and answer questions collaboratively.

Accommodations

The University of Toronto supports accommodations for students with diverse learning needs, which may be associated with mental health conditions, learning disabilities, autism spectrum, ADHD, mobility impairments, functional/fine motor impairments, concussion or head injury, visual impairments, chronic health conditions, addictions, D/deaf, deafened or hard of hearing, communication disorders and/or temporary disabilities, such as fractures and severe sprains, or recovery from an operation.

If you have a learning need requiring an accommodation the University of Toronto recommends that students [register with Accessibility Services](#) [↗] as soon as possible.

We know that many students may be hesitant to reach out to Accessibility Services for accommodations. The process of accommodation is private; we will not share details of your needs or condition with any instructor.

If you feel hesitant to register with us, we encourage you to reach out for further information and resources on how we can support. It may feel difficult to ask for help, but it can make all the difference during your time here.

Phone: 416-978-8060

Email: accessibility.services@utoronto.ca

Equity, Diversity and Inclusion

U of T Engineering strives to create equitable and inclusive learning environments for all individuals. Each person has their own lived experiences and socio-cultural identities that shape their worldview and their experiences with privilege and oppression.

Looking for community? Feeling isolated? Not being understood or heard?

You are not alone. You can talk to anyone in the Faculty that you feel comfortable approaching, anytime — professors, instructors, teaching assistants, [department academic advisors](#) [↗], student leaders or the [Assistant Dean of Diversity, Inclusion and Professionalism](#).

You belong here. In this class, the participation and perspectives of everyone is invited and encouraged. The broad range of identities and the intersections of those identities are valued and create an inclusive team environment that will help you achieve academic success. You can read the evidence for this approach [here](#) [↗].

You have rights. The [University Code of Student Conduct](#) [↗] and the [Ontario Human Rights Code](#) [↗] protect you against all forms of harassment or discrimination, including but not limited to acts of racism, sexism, Islamophobia, antisemitism, homophobia, transphobia, ableism, classism and ageism. Engineering denounces unprofessionalism

or intolerance in language, actions or interactions, in person or online, on- or off-campus. Engineering takes these concerns extremely seriously and you can confidentially disclose directly to the Assistant Dean for help [here](#).

Support for Indigenous Students

If you are an Indigenous engineering student, you are invited to join a private Discord channel to meet other Indigenous students, professors, and staff, chat about scholarships, awards, work opportunities, Indigenous-related events, and receive mentorship. Email [Professor Jason Bazylak](#) if you are interested.

Indigenous students at U of T are also invited to visit Nations House's (FNH) Indigenous Student Services for culturally relevant programs and services. If you want more information on how to apply for Indigenous specific funding opportunities, cultural programs, traditional medicines, academic support, monthly social events or receive the weekly newsletter, go to the FNH [website](#) [↗], [email](#) or follow FNH on social media: [Facebook](#) [↗], [Instagram](#) [↗], or [TikTok](#) [↗]. A full event calendar is on the CLNX platform. Check CLNX often to see what new events are added!

As part of the Faculty's commitment to improving Indigenous inclusion, all U of T Engineering faculty, staff and students are called upon to start or continue their personal journeys towards understanding and acknowledging Indigenous peoples' history, truths and culture. The Faculty's 2018 [Blueprint for Action report](#) [↗] identified many actions to (re)building relationships between U of T Engineering and Indigenous peoples. These actions also align with the Truth and Reconciliation Commission of Canada's [Calls to Action](#) [↗].

One action is for non-Indigenous people to make a Land Acknowledgement in lectures and/or before important meetings. For reference, here are examples from the University and the COU: [University of Toronto's Statement of Acknowledgement of Traditional Land](#) [↗] and [Council of Ontario Universities' Acknowledgement of Traditional Land](#) [↗].

To combat the fear that land acknowledgements have become largely performative and rote, we advise that the acknowledgement is done with sincerity and reflection on how you benefit from using the land, and on your commitments to Truth and Reconciliation. Take this [training](#) [↗] to learn more.

As an environmentalist, I have always been appreciative of the fact that right now we are in the territory covered by the Dish With One Spoon Wampum Belt. This is a treaty originally by the Anishinaabe, Mississaugas, and the Haudenosaunee. It is not a rights-based agreement, but rather a responsibility-based agreement. Hundreds of years ago the First Nations People in this area recognized our responsibility to share and protect the land. Since that first wampum belt many other First Nations and Europeans have joined in this treaty. The First Nations close ties to the land enabled them to see the needs of the land when others could not or choose not to see. For that I am thankful to the Indigenous People of this land we work and study on.

Looking for help personalizing the land acknowledgement? [Professor Jason Bazylak](#) [↗] is the Dean's Advisor on Indigenous Inclusivity. He is willing to chat with anyone on how to go about presenting and/or discussing this with your class. He can be reached at jason.bazylak@utoronto.ca. You can also contact Marisa Sterling, P.Eng. in the Dean's Office at marisa.sterling@utoronto.ca.

Wellness and Mental Health Support

As a University of Toronto Engineering student, you have a Departmental [Undergraduate Advisor](#) [↗] or a Departmental [Graduate Administrator](#) [↗] who can support you by advising on personal matters that impact your academics. Other resources that you may find helpful are listed on the [U of T Engineering Mental Health & Wellness webpage](#) [↗], and a small selection are also included here:

- [U of T Engineering's Student & Community Wellness Coordinator](#) [↗]
- [Health & Wellness](#) [↗] and the [On-Location Engineering Wellness Counsellor](#) [↗]
- [Health & Wellness Peer Support Program](#) [↗]
- [Accessibility Services](#) [↗] & the [On-Location Advisor](#) [↗]
- [Graduate Engineering Council of Students' Mental Wellness Commission](#) [↗]
- [SKULE™ Mental Wellness](#) [↗]
- [U of T Engineering's Learning Strategist](#) [↗] and [Centre for Learning Strategy Support](#) [↗]
- [Registrar's Office](#) [↗] and [Scholarships & Financial Aid Office & Advisor](#) [↗]

We encourage you to access these resources as soon as you feel you need support; no issue is too small. You may reach out to the counsellors at [U of T Telus Health Student Support](#) [↗] for 24/7 free and confidential counselling support.

If you find yourself feeling distressed and in need of more immediate support visit uoft.me/feelingdistressed [↗] or U of T Engineering's [Urgent Support – Talk to Someone Right Now](#) [↗].

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